

19 Nitrogen compounds

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Amines are bases because the nitrogen lone pair accepts H^+ ; electron donation by alkyl groups can alter basicity. Amines react with acids to form ammonium salts and with acyl chlorides to form amides. Amino acids have both amino and carboxyl groups and often exist as zwitterions depending on pH.

Worked example: follow the reasoning

Ethylamine + $HCl \rightarrow$ ethylammonium chloride; the amine acts as Brønsted base.

Quick check · multiple choice

A What acts as a Brønsted base in amines?

[1]

- A N-H bond only
- B Carbon skeleton
- C Nitrogen lone pair
- D Carboxyl oxygen always

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain why glycine can migrate toward a cathode at sufficiently low pH and show its predominant charged form. [3]

Answers and explanation

Quick check: C. Nitrogen lone pair

Question 1 [3 marks]

At low pH amino group is protonated [1] and carboxyl group mainly COOH [1]; $\text{H}_3\text{N}^+\text{CH}_2\text{COOH}$ has net + charge and moves toward cathode [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.